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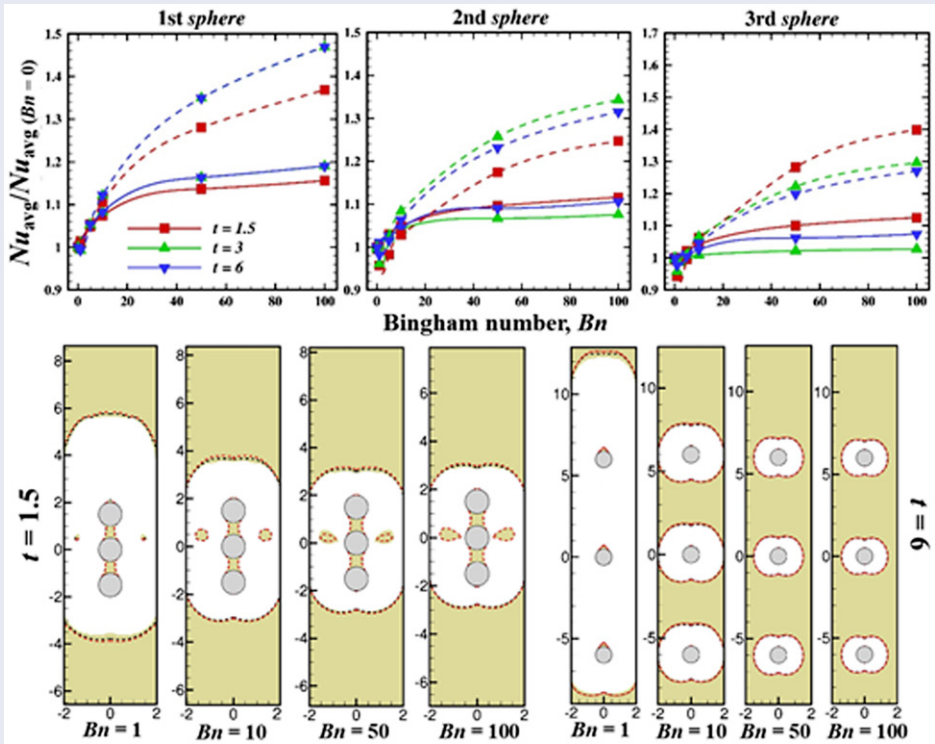
Nabankur Dasgupta[a,b], Rituraj Borah[a], Pragya Mishra[a], Anoop Kumar Gupta[a,c], and Rajendra Prasad Chhabra[a]

[a]Department of Chemical Engineering, Indian Institute of Technology Kanpur, Kanpur, India; [b]Department of Engineering Science and Mechanics, The Pennsylvania State University, University Park, PA, USA; [c]Department of Civil & Environmental Engineering, National University of Singapore, 1 Engineering Drive 2, Singapore

ABSTRACT

This work reports results on the drag and heat transfer from an in-line array of three isothermal spheres falling in a cylindrical confinement filled with Bingham plastic fluids. The effects of dimensionless parameters, such as the Reynolds number ($1 \leq Re \leq 100$), Prandtl number ($1 \leq Pr \leq 100$), Bingham number ($0 \leq Bn \leq 100$), blockage ratio ($2 \leq \beta \leq 4$) and sphere-to-sphere distance ($1.5 \leq t \leq 6$) have been elucidated. The flow and heat transfer characteristics were analysed in terms of yielded/unyielded regions, streamline and isotherm contours, drag coefficient, pressure coefficient, and local and average Nusselt number. Broadly, the drag coefficient shows a positive dependence on *Bn* and sphere-to-sphere distance (*t*) while it exhibits an inverse dependence on *Re* and β. On the other hand, the Nusselt number shows a positive dependence on *Re, Pr, Bn* and β; and a complex dependence on *t* for each sphere. Simple predictive expressions for the average Nusselt number for each sphere are formulated, thereby enabling its prediction in a new application.

GRAPHICAL ABSTRACT



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CONTACT Rajendra Prasad Chhabra chhabra@iitk.ac.in Indian Institute of Technology Ropar, Chemical Engineering, Nangal Road, Rupnagar 140001, India.
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1. Introduction

The flow of non-Newtonian fluids past and heat transfer from single and multiple spheres denotes an idealization of several industrially important processes. Typical examples include packed and fluidized bed reactors to carry out polymerization reactions, heating, cooling and transportation of particulate slurries encountered in mineral, food, pharmaceutical sectors, etc.[1–4] Furthermore, reliable knowledge of the hydrodynamic behaviour of single and multiple particles in different types of fluids is also central for formulating constitutive equations to describe the rheological behaviour of suspensions.[4,5] In addition to these wide-ranging potential applications, such model studies entailing single and in-line arrays of multiple spheres also serve as a launching pad to understand the nature of inter-particle interactions in concentrated particulate systems like dispersions and slurries. Additional examples are found in the flow of blood in arteries in the form of a chain of red blood cells, and in narrow chromatographic separation columns. In addition to the physical properties of the solid-fluid system, the momentum and heat transfer characteristics of even a single sphere are also strongly influenced by the presence of confining walls. Thus, for instance, in the simplest case of a single sphere settling at the axis of a cylindrical tube filled with a fluid, not only the drag on the sphere is augmented (thereby slowing down its sedimentation) due to the extra energy dissipation at the walls, but the rate of heat transfer between the sphere and the fluid (or vice versa) is also influenced due to the sharpening of the temperature gradient on the surface of the sphere. Additional effects also arise due to hydrodynamic interactions in the case of multiple spheres depending upon their geometrical configuration like their arrangement and gap, etc. Consequently, over the years, much has been written about these and the other related effects on the momentum and heat transfer aspects of single and arrays of spheres in Newtonian fluids. This body of literature has been reviewed, amongst others, by Michaelides,[6] Maheshwari et al.,[7] and others.[8–10] All in all, an adequate body of knowledge is now available on these aspects in Newtonian fluids.

Similarly, a reasonable body of information, though less extensive than that for Newtonian fluids, is available on the drag and heat transfer characteristics of a single sphere in unconfined and confined power-law and Bingham plastic fluids, e.g., see refs.[3,11–20] In power-law fluids, both drag and Nusselt number increase above the corresponding values for an unconfined sphere. However, the severity of this effect is strongly modulated by the degree of interaction between the boundary layers developed on the sphere and on the confining walls. The boundary layer thickness itself is determined by an intricate interplay between the blockage ratio (β), power-law index (n) and Reynolds number (Re). Therefore, the degree of enhancement in drag and Nusselt number for a single sphere shows non-monotonous trends with respect to these variables.[14–20] In contrast, in Bingham plastic fluids, additional complications arise due to the co-existence of fluid-like and solid-like domains depending upon the magnitude of the local stress *vis-à-vis* the fluid yield stress. Thus, if the fluid-like region encapsulating the sphere does not intersect with the physical tube wall, the sphere does not see the wall so to say and there is no effect of confinement in this case. Conversely, if the fluid-like region extends up to the tube wall, stresses are transmitted through the yielded fluid and the walls exert an extra retardation effect on the sedimentation of a single sphere.[19–23] Overall, there exists a fair degree of match between the predicted and experimental values of drag (and to a limited extent of Nusselt number) for a single sphere in confined and unconfined power-law fluids, e.g., see refs.[3,24] On the other hand, while drag values in unconfined Bingham plastic fluids show good correspondence with the available experiments,[3,25,26] only limited results are available on wall effects and no heat transfer data are currently available to corroborate the numerical predictions.[25,26] Evidently, this body of knowledge is neither as extensive nor as coherent as that for Newtonian fluids.

The next generation of studies in this field have dealt with the behaviour of arrays of multiple spheres (up to 8) in Newtonian fluids.[7–10] However, only scant results are available on two or more spheres in power-law and Bingham plastic fluids.[3] In the few experimental studies dealing with two or more spheres in power-law fluids,[27–29] the major thrust has been on the segregation of spheres in the free falling regime to understand the formation of clusters in the hindered settling of concentrated suspensions and/or on wall effects.[30,31] Scant results in viscoelastic fluids[30] indicate that the two in-line spheres may drift away or approach each other depending upon the initial separation and the celerity of viscoelastic effects. In an extensive study, Zhu et al.[29] measured drag forces in power-law fluids on multiple spheres using a micro force balance. They also supplemented their experimental results by numerically simulating the flow over two spheres (separated by varying gaps ranging from 0 to 2 times the sphere diameter) in mildly shear-thinning fluids ($0.6 < n < 1$) over the range of Reynolds number of 0.7 to 23. However, no corresponding results are available on heat transfer in these systems. In contrast, the interaction between spheres separated by varying gaps is expected to be more complex in the case of Bingham plastic fluids. Thus, for instance, the fluid-like cavity encapsulating a moving sphere is of the order of 2-3 times the sphere diameter.[11,22] It thus stands to reason that if the two spheres are separated by a distance larger than 2-3d, they are more likely to behave as non-interacting spheres. On the other hand, if the trailing sphere is well within the yielded region of the leading sphere, there will be hydrodynamic interactions between the two. Indeed, based on the numerical results (creeping regime) for two co-axial spheres in Bingham plastic fluids, Liu et al.[32] identified three regimes. For a fixed value of the Bingham number; spheres moving in two independent yielded regions ($L/R > 5.5$); the spheres moving without touching each other in the same yielded region ($4.5 \leq L/R \leq 5.5$) and the two spheres joined by unyielded material bridge ($L/R \leq 4.5$). Similarly, Jie and Zhu[33] computed the drag force for arrays of spheres containing up to 7 spheres in Herschel-Bulkley fluids. They also observed trends qualitatively similar to that of Liu et al.[32] Based on the preceding discussion, it is thus fair to conclude

that the few studies dealing with two or more spheres in Bingham plastic fluids are not only limited to the creeping flow regime but also to the momentum transfer aspects only. Hence, no prior results are available on forced convection heat transfer in Bingham plastic fluids from an array of three heated spheres. This work endeavours to fill this gap in the literature by numerically investigating the drag and Nusselt number behaviour for an array of three isothermal spheres fixed at the axis of a long tube filled with a Bingham fluid medium thereby elucidating the combined effects of blockage and fluid yield stress on the hydrodynamic and heat transfer phenomena.

In particular, the momentum and energy equations have been solved numerically to obtain the momentum and heat transfer characteristics for an in-line array of three heated spheres for the following ranges of conditions: $1 \leq Re \leq 100$, $1 \leq Pr \leq 100$, $1 \leq Bn \leq 100$, $1.5 \leq t \leq 6$ and $2 \leq \beta \leq 4$ thereby delineating the combined effects of blockage (β), gap between spheres (t) and the fluid yield stress (Bn). The present results are compared with the previous studies as much as possible.

2. Problem formulation and governing equations

The sedimentation of a chain of three spheres separated by a fixed gap is simulated here by having the spheres fixed at the axis of a tube and letting the fluid-tube assembly move with a uniform velocity U_0, as shown schematically in Figure 1(a). Each sphere, heated to T_w, is of diameter d and separated by a center-to-center distance of s in a tube of diameter D. The fluid enters the tube at a lower temperature of T_o, as shown in Figure 1(a). In order to exploit the symmetry of the flow, the computational domain is shown in Figure 1(b), along with the coordinate system used here. The thermo-physical properties of the fluid are assumed to be independent of the temperature and the viscous dissipation term in the energy equation is also neglected thereby leading to the decoupling of the velocity and temperature fields. The assumption of the constant thermo-physical properties limits the applicability of the present results to situations for small values of the temperature difference $\Delta T = T_w - T_o$, which is only 5K in the present case. The sphere diameter d and the uniform inlet velocity U_0 are used to scale the spatial variables and velocities respectively. In the energy equation, temperature was made dimensionless as $T = (T' - T_o)/(T_w - T_o)$. With these simplifying assumptions and scaling considerations, the continuity, momentum and the energy equations are written in their dimensionless forms as:

Continuity equation:

$$\nabla . V = 0 \qquad [1]$$

Momentum equation:

$$(V.\nabla)V = -\nabla P + \frac{1}{Re}\nabla.\tau \qquad [2]$$

Energy equation:

$$(V.\nabla)T = \frac{1}{Re \cdot Pr}\nabla^2 T \qquad [3]$$

The extra stress tensor for an incompressible fluid is given as follows:

$$\tau = \eta\dot{\gamma} \qquad [4]$$

Here, η is the scalar viscosity function and $\dot{\gamma}$ is the rate of deformation tensor, which can be expressed in terms of the velocity gradients:

$$\dot{\gamma} = (\nabla V + \nabla V^T) \qquad [5]$$

The magnitudes of the extra stress tensor and the rate of deformation tensor are given as follows:

$$|\tau| = \sqrt{tr(\tau^2)} \qquad [6]$$

$$|\dot{\gamma}| = \sqrt{tr(\dot{\gamma}^2)} \qquad [7]$$

For a Bingham plastic fluid, the dimensionless constitutive equation is written as:

$$\tau = \left(1 + \frac{Bn}{|\dot{\gamma}|}\right)\dot{\gamma}, \quad \text{if} |\tau| > Bn \qquad [8a]$$

$$\dot{\gamma} = 0, \quad \text{if} |\tau| \leq Bn \qquad [8b]$$

Here $\dot{\gamma}$ and τ are non-dimensionalised by using U_0/d and $\mu_B U_0/d$ as characteristic scales. The problem closure is obtained by identifying the physically realistic boundary conditions for this case as follows:

- No-slip and constant temperature conditions are used on the surface of the sphere:

$$V_z = 0, \; V_r = 0 \text{ and } T = 1.$$

- Uniform axial velocity and temperature at the inlet of the tube:

$$V_z = 1, \; V_r = 0 \text{ and } T = 0 \text{ for } 0 \leq r \leq \beta/2.$$

- The moving wall is assumed to be adiabatic:

$$V_z = 1, \; V_r = 0 \text{ and } \frac{\partial T}{\partial r} = 0 \text{ at } r = \beta/2.$$

- At the exit of the tube, the standard outflow condition, i.e., $\partial\phi/\partial z = 0$ where $\phi = V_r \text{or} V_z \text{or} T$. However, the gradients in the radial direction may still exist. This is similar to the Neumann type boundary condition commonly used in such situations. In addition, in order to economize on the computational effort, symmetry conditions ($V_r = 0$; $\partial V_z/\partial r = 0$ and $\partial T/\partial r = 0$) were used at the axis of the tube.

An examination of the governing differential equations along with the boundary conditions outlined above suggests that the flow field (momentum transfer aspects) is influenced by four dimensionless groups namely, Reynolds number (Re), Bingham number (Bn), blockage ratio (β) and separation between the spheres ($t = s/d$) whereas the temperature field (and hence heat transfer) shows additional dependence on the Prandtl number (Pr). This work aims to explore and develop these functional relationships. Also, it is

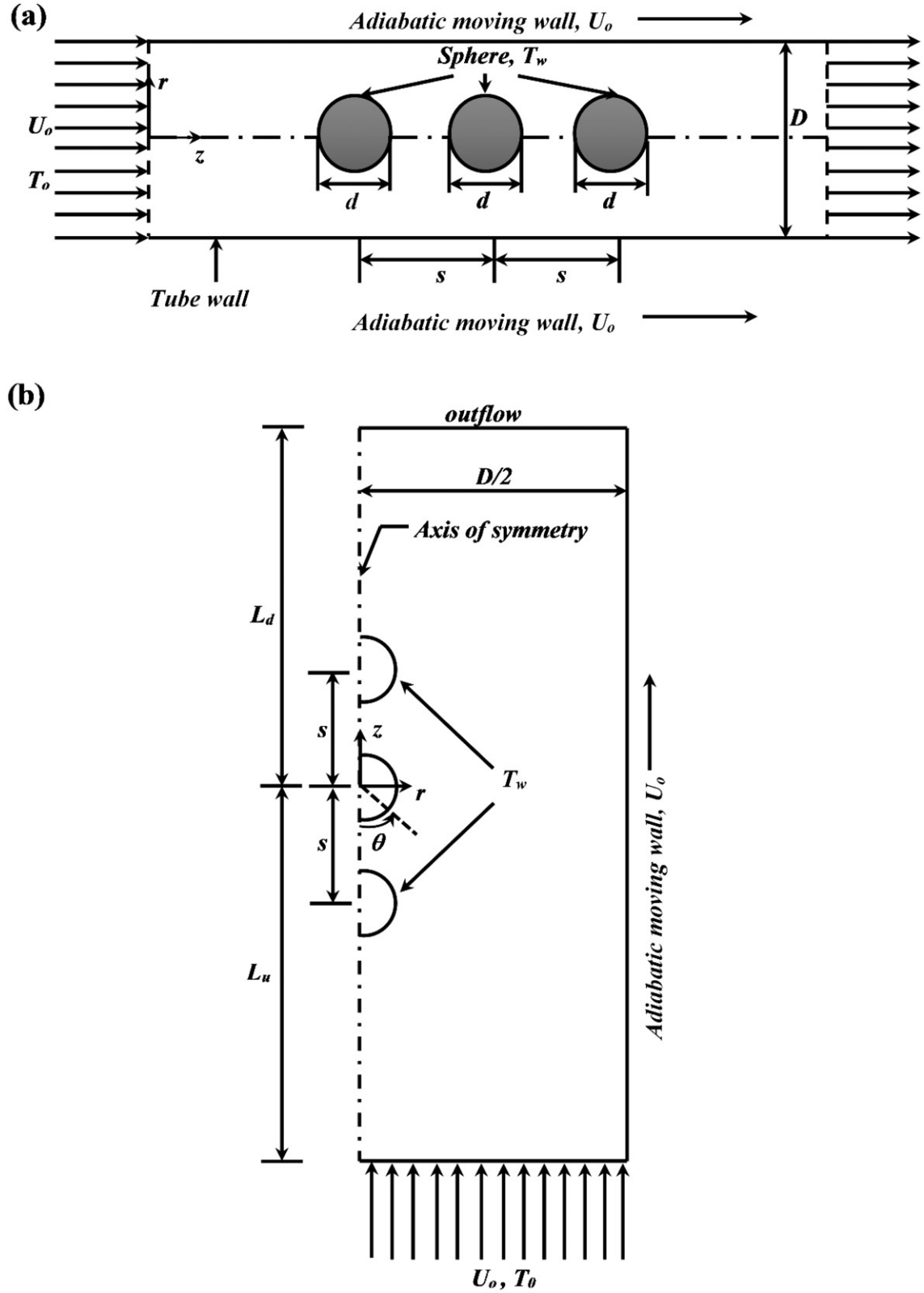


Figure 1. (a) Schematics of the flow (b) computational domain.

useful to define some of these dimensionless parameters here as:

Bingham number:

$$Bn = \frac{\tau_o d}{\mu_B U_0} \quad [9]$$

Reynolds number:

$$Re = \frac{\rho d U_o}{\mu_B} \quad [10]$$

Prandtl number:

$$Pr = \frac{C\mu_B}{k} \quad [11]$$

In order to ascertain the relative importance of conduction and convection, it is customary to use the Peclet number, $Pe = Re.Pr$, which is, however, not an independent group. The preceding definitions of the Reynolds and Prandtl numbers are based on the choice of μ_B being the representative fluid viscosity. This, of course, does not incorporate the effect of yield stress. Therefore, it has been argued and practiced in the literature[25,26,34] to introduce an effective viscosity given as $(\mu_B + \tau_0 d/U_0)$ being more appropriate than μ_B on its own. Such rescaling leads to the following modified definitions of the Reynolds (Re^*) and Prandtl (Pr^*) numbers:

$$Re^* = \frac{Re}{(1+Bn)} \quad [12]$$

$$Pr^* = Pr(1+Bn) \quad [13]$$

Two observations are in order here: In the limit of $Bn \to 0$, both these definitions reduce to their expected forms for Newtonian fluids and hence this approach offers the possibility of consolidating the results for Newtonian and Bingham plastic fluids. Secondly, this rescaling does not change the definition of the Peclet number because it is independent of the fluid viscosity. This approach thus affords the possibility of reconciling heat transfer results for the Newtonian and Bingham plastic fluids. Both these ideas are explored in this work.

3. Numerical solution methodology and the choice of numerical parameters

The numerical solution of the governing equations maps the flow domain in terms of the velocity, pressure and temperature fields which can be post processed to deduce the values of the global quantities like drag coefficient, Nusselt number, etc., as detailed in some recent studies.[25,26] Therefore, only key details about the computational mesh and other aspects of the numerical solution method used are discussed here. However, prior to undertaking the detailed discussion of these aspects, it is useful to make some observations about the intrinsic difficulties in incorporating the Bingham model, Equation (8), into a numerical solution scheme due to its discontinuous form. In order to obviate this difficulty, regularization methods have been developed which effectively convert the abrupt transition between the fluid-like and solid-like regions into a gradual one.[35–38] The three commonly used regularization approaches are introduced here:

Papanastasiou model: [37]

$$\tau = \left(1 + \frac{Bn\{1-\exp(-m|\dot{\gamma}|)\}}{|\dot{\gamma}|}\right)\dot{\gamma} \quad [14]$$

and the scalar viscosity η is given as:

$$\eta = \left(1 + \frac{Bn\{1-\exp(-m|\dot{\gamma}|)\}}{|\dot{\gamma}|}\right) \quad [15]$$

Naturally, the two extreme conditions $Bn \to 0$ and $Bn \to \infty$ correspond to the Newtonian fluid and fully plastic flow regime, respectively. For high values of m, Equation (14) approaches the true Bingham plastic model. However, excessively large values of m lead to numerical instabilities due to the stiffness of the coefficient matrix and hence an appropriate value of m must be chosen to strike a balance between the precision of the results on one hand and the numerical challenges on the other.

Bi-viscous model:[36]

$$\eta = \frac{\mu_y}{\mu_B}, \quad \text{for} |\tau| < Bn \quad [16]$$

$$\eta = 1 + Bn\left\{1 - \frac{1}{(\mu_y/\mu_B)}\right\}\frac{1}{|\dot{\gamma}|}, \quad \text{for} |\tau| \geq Bn \quad [17]$$

In this approach, the unyielded regions are replaced by a fluid of very high viscosity ($\mu_y \gg \mu_B$). Naturally, $\mu_y \to \infty$ corresponds to the true Bingham behaviour and hence large values of μ_y must be employed which is often several orders of magnitudes greater than the plastic viscosity (μ_B). Once again, excessively large values can also lead to numerical difficulties.

Bercovier and Engelman model:[35]

$$\eta = 1 + Bn\left(\frac{1}{|\dot{\gamma}| + \lambda}\right) \quad [18]$$

This approach differs from the first two in so far that it permits a vanishingly small value of strain rate, λ, in the unyielded regions as opposed to $\lambda = 0$ in a true Bingham model fluid. Nonetheless, the success of these three approaches hinges on a prudent choice of m or μ_y or λ. But none of these approaches resolve the stress field entirely satisfactorily, albeit this weakness does not seem to influence the values of the gross engineering parameters like drag coefficient and Nusselt number. On the other hand, it is fair to add here that one can use more advanced techniques like operator-splitting methods, augmented Lagrangian methods, or those based on the variational inequalities, etc. to circumvent the need for a regularization method.[39–42] Since the shape and locations of the yield surfaces (separating the fluid-like and solid-like region) are not known *a priori*, these methods tend to be far more computation intensive than those based on the use of the regularized methods. Also, the limited results obtained using the two approaches for a few cases are in fair agreement thereby inspiring confidence in the validity of the regularized form of the Bingham model.[41]

In this study, the governing equations subjected to the given boundary conditions together with the constitutive relations are solved using the finite element-based solver COMSOL Multiphysics® (version 4.3a). A mesh using quadrilateral cells has been implemented to solve the

discretized form of the governing equations. At high values of the Reynolds number, Prandtl number and Bingham number, the velocity and temperature gradients are expected to be steep at the solid walls and near the yield surfaces separating the solid-like and fluid-like regions and hence a relatively finer mesh has been implemented in these regions. A relative convergence criterion of 10^{-5} has been used for the momentum and energy equations. The yield surfaces are located by employing the von Mises yielding criterion [43] with the relative tolerance of 10^{-6}.

As it is simply not possible to consider an infinitely long tube, choice of the suitable values of L_u and L_d (Figure 1(b)) along with an appropriate numerical mesh is thus necessary to strike a balance between the requisite computational resources and the precision of the results. For this purpose, several values of $L_u = 75$, 100, 125 and $L_d = 100$, 125, 150 were tested for the values of $Re = 1$, $Pr = 1$ and $Bn = 0$ (Newtonian fluid behaviour) used in this work, for both velocity and temperature fields exhibit slow spatial decay under these conditions. Based on a detailed examination of the values of the gross parameters (Nu_{avg}, C_D and C_{DP}), the values of $L_u = 100$ and $L_d = 125$ were found to be quite adequate and these are superior than the values used in the previous studies on a single sphere.[12,21] Similarly, to capture the steep gradients at high Reynolds number and Prandtl number, three quadrilateral non-uniform grids (*G1*, *G2*, *G3*) were created for each configuration with adequate refinements of the element size. Figure 2 shows typical grid effects on the local Nusselt number for a range of conditions. Evidently, the results obtained using grids *G2* (∼ 80,000 CVs) and *G3* (∼ 150,000 CVs) are virtually indistinguishable from each other and hence *G2* was chosen as the final grid used in this study; it consists of total ∼80,000 CVs in the computational domain, 180 grid points on the surface of each sphere, smallest element size ∼0.008*d* and with ∼310,000 degree of freedoms. Similarly, an optimum value

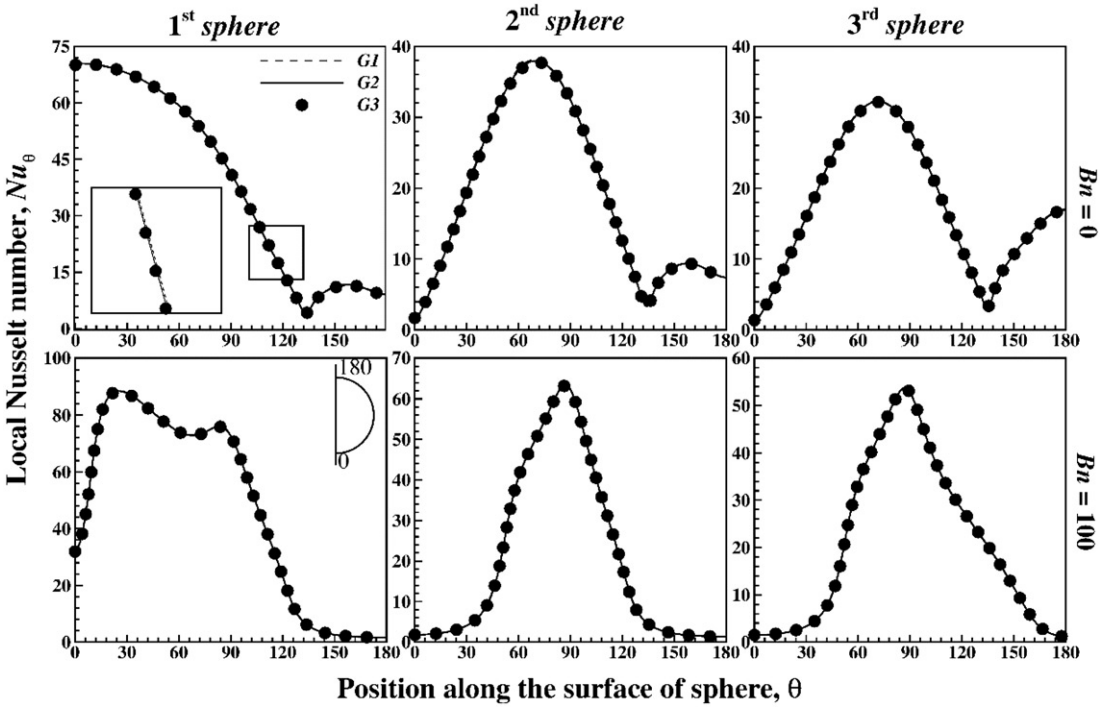


Figure 2. Grid test at $Re = 100$ and $Pr = 100$ for $t = 1.5$, $\beta = 2$.

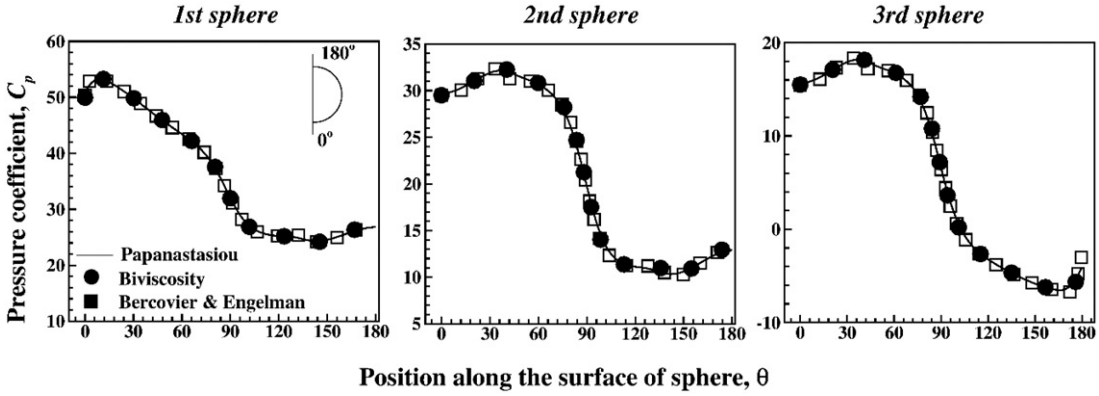


Figure 3. Comparison of three regularization techniques ($m = 10^5$, $\mu_y/\mu_B = 10^4$ and $\lambda = 10^{-10}$) in terms of the surface pressure coefficient over the sphere surface at $Re = 100$ and $Bn = 100$ for $t = 1.5$,$\beta = 2$.

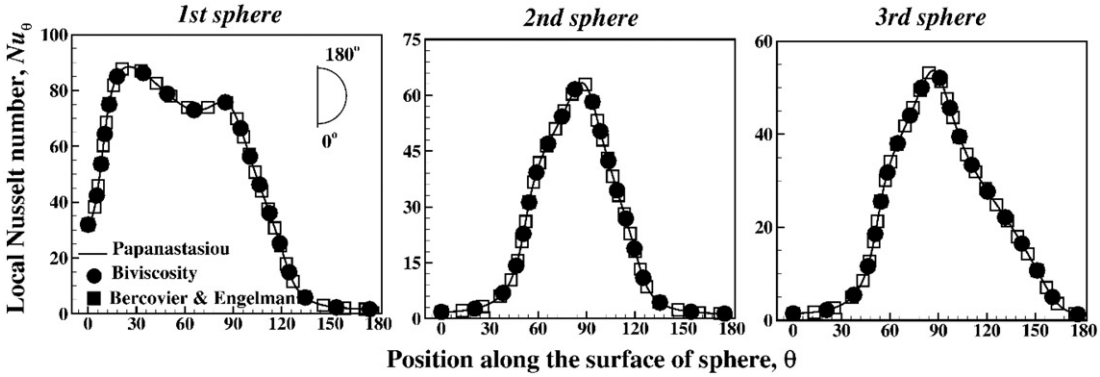


Figure 4. Comparison of three regularization techniques ($m = 10^5$, $\mu_y/\mu_B = 10^4$ and $\lambda = 10^{-10}$) in terms of the local Nusselt number over the sphere surface at $Re = 100$, $Pr = 100$ and $Bn = 100$ for $t = 1.5$, $\beta = 2$.

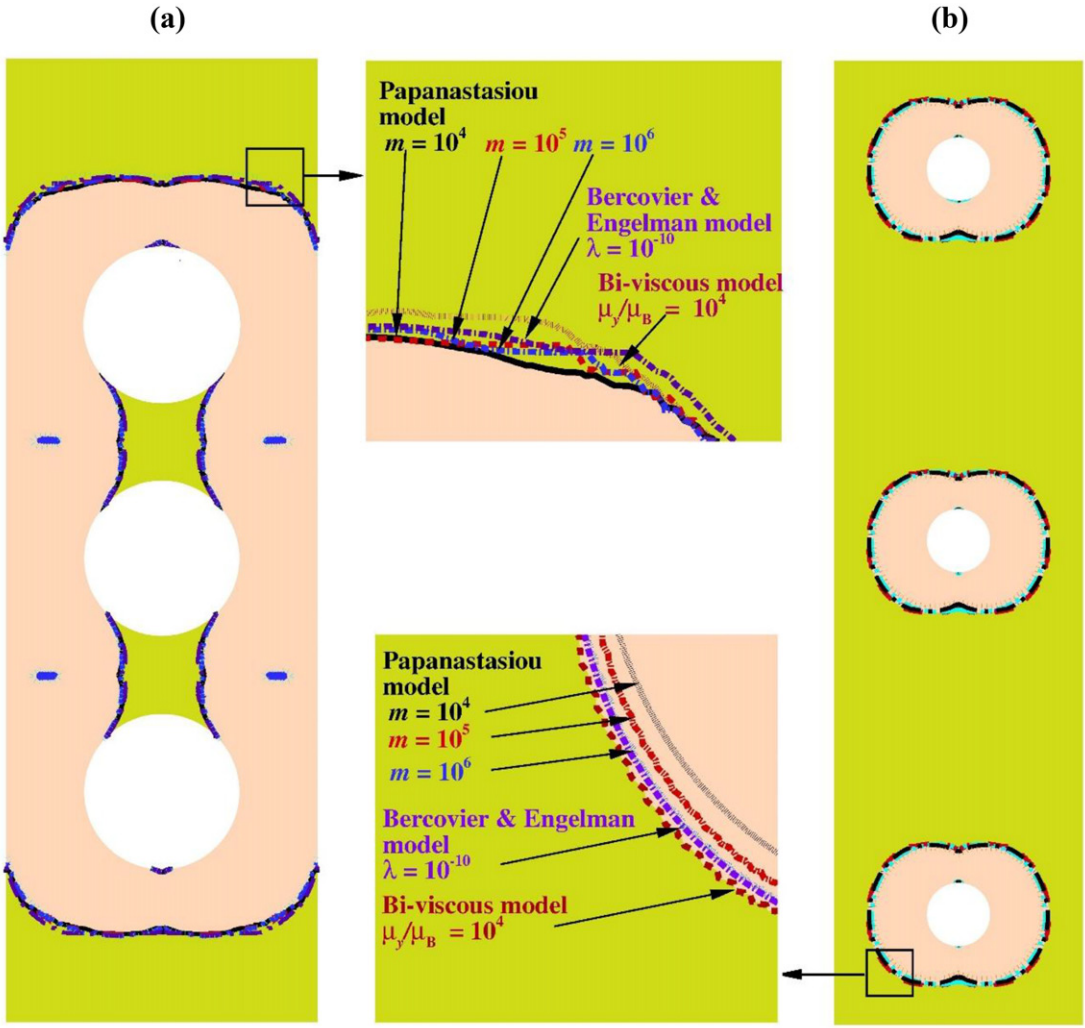


Figure 5. Comparison of yield surfaces for three regularization schemes at $Re = 100$, $Pr = 100$ and $Bn = 100$ for (a) $t = 1.5$, $\beta = 2$ (b) $t = 6$, $\beta = 4$.

of the regularization parameter m or μ_y or λ must also be chosen to obtain results free from the numerical artifacts stemming from the regularization approximations. Figures 3 and 4 show comparisons between the predictions of the three regularization schemes in terms of the pressure coefficient and local Nusselt number respectively. The predictions of the yield surfaces based on the three regularization methods are presented in Figure 5. In view of such a close correspondence between these predictions seen in Figures 3-5, most of the results reported in this work are based on the application of the Papanastasiou model with $m = 10^5$. Further justification of the numerical parameters used in this work is presented in the next section by way of presenting a few benchmark comparisons.

4. Results and discussion

The new extensive results for an array of three spheres in Bingham fluids in a cylindrical tube presented here extend over the following ranges of conditions: $1 \leq Re \leq 100$, $1 \leq Pr \leq 100$, $0 \leq Bn \leq 100$, $1.5 \leq t \leq 6$ and $2 \leq \beta \leq 4$. The influence of each of these parameters has been illustrated with the help of streamline and isotherm contours, yield surfaces, pressure coefficient, drag coefficient and Nusselt number for each sphere. However, before undertaking a detailed discussion of the new results, it is instructive to validate the present solution methodology.

4.1. Validation of results

Ramachandran et al.[9] have reported numerical results for the forced convection heat transfer of interacting spheres in air and Figure 6 shows a comparison between their and the present results for a linear array of three heated spheres with variable separation between the 1st and 2nd (t_{12}) and between the 2nd and 3rd (t_{23}) spheres without the confining walls, i.e., for $\beta = \infty$. An excellent match is seen to exist between the two sets of results. Next, the present results are compared with that of Lloyd and Boehm[8] in terms of the drag coefficient and the average Nusselt number for an in-line array of eight spheres at $Re = 40$ (Table 1). The two heat transfer results are generally within 2% of each other whereas the drag values differ by 5-6%, especially for the fourth sphere onwards. While the exact reasons for these differences are not immediately obvious, suffice to add here that the grid and domain used here are much superior than that used by Lloyd and Boehm[8] and the differences of this order are not uncommon in such studies.[44] Besides, a well-developed wake forms in the rear of a single sphere at about $Re \sim 22$-24 and therefore, at $Re = 40$, it is likely that the wake of the preceding sphere extends up to the succeeding sphere. An adequate numerical resolution is thus needed of the flow in this region to obtain reliable results. Suffice it to add here that since the numerical mesh used here is much finer than that in the study of Lloyd and Boehm,[8] the present results are regarded to be more reliable. Similarly, Table 2 shows a comparison with the results (obtained using the finite volume based Fluent) of Maheshwari et al.[7] for three spheres at the axis of a tube for Newtonian fluids. Notwithstanding the fact that different numerical solution methodologies have been used in the two studies, the two predictions are in excellent agreement. Finally, limited results were also obtained in this work for single sphere and two-sphere systems undergoing steady translation in unconfined Bingham plastic fluids in order to validate the present solution methodology with the creeping flow results of Liu et al.[32,45] and Jie and Zhu,[33] Figure 7(a) and 7(b). In both cases, an excellent agreement is seen to exist. Based on the preceding discussion, the new results reported herein are regarded to be reliable within 1.5–2%.

4.2. Yielded/unyielded zones

One of the key features which sets a Bingham plastic fluid apart from a Newtonian fluid is that the former is not yielded everywhere in the flow domain. The so-called fluid-like (yielded) and solid-like (unyielded) regions are separated by yield surfaces, i.e., in the regions where the magnitude of the stress tensor is below that of the yield stress. Owing to the regularization method employed here, these are approximated here by the constant stress contours, which denote stress values within 10^{-4}% of the fluid yield stress.

Figure 8 shows typical yield surfaces for a range of conditions. Intuitively, it appears that the fluid should yield near the solid boundaries (due to the imposition of the no-slip condition) and in the regions where the velocity profile is

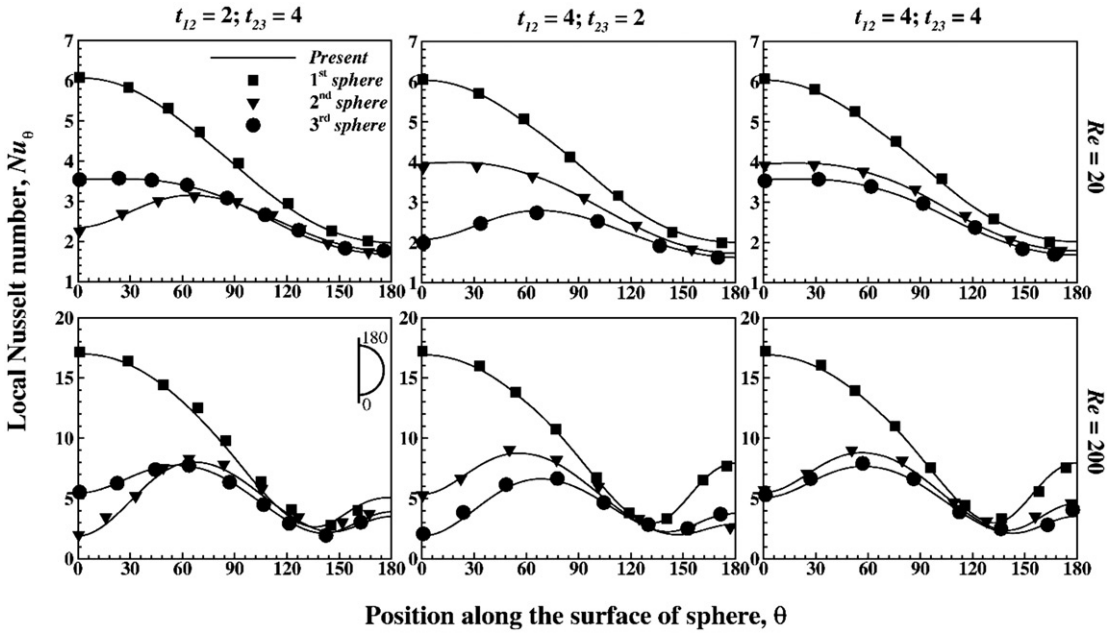


Figure 6. Comparison of present results (lines) for the local Nusselt number for an unconfined flow of Newtonian fluids with that from ref. [9] (symbols) at $Pr = 0.7$.

Table 1. Comparison of present results with Lloyd and Boehm[8] at $Re = 40$ (Newtonian fluid).

	C_D		Nu_{avg} ($Pr = 0.73$)		Nu_{avg} ($Pr = 7.3$)	
Sphere No.	Lloyd and Boehm[8]	Present	Lloyd and Boehm[8]	Present	Lloyd and Boehm[8]	Present
1	1.768	1.8	5.0119	4.917	9.872	9.686
2	0.99	1.003	5.651	5.712	11.829	12.046
3	0.872	0.908	5.664	5.751	12.327	12.553
4	0.799	0.837	5.642	5.721	12.362	12.599
5	0.751	0.807	5.586	5.619	12.265	12.509
6	0.717	0.759	5.564	5.602	12.119	12.364
7	0.699	0.727	5.542	5.579	12.18	12.290
8	0.71	0.751	5.555	5.643	12.201	12.396

Table 2. Comparison of the present results with Maheshwari et al.[7] (a) drag coefficient (b) average Nusselt number at $Pr = 0.74$ (air).

(a) Drag coefficient (C_D)

	1st sphere		2nd sphere		3rd sphere	
Re	Present	ref. [7]	Present	ref. [7]	Present	ref. [7]
$\beta = 2$, $t = 2$						
5	28.74	28.67	28.70	28.65	28.76	28.73
10	14.56	14.53	14.50	14.48	14.55	14.53
20	7.61	7.59	7.48	7.47	7.51	7.51
50	3.62	3.61	3.30	3.29	3.32	3.31
100	2.32	2.32	1.86	1.85	1.86	1.85
$\beta = 6$, $t = 4$						
5	8.18	8.18	7.62	7.62	7.67	7.67
10	4.72	4.71	4.03	4.02	4.03	4.03
20	2.89	2.89	2.23	2.23	2.17	2.17
50	1.64	1.64	1.11	1.11	1.03	1.03
100	1.12	1.12	0.69	0.69	0.63	0.63

(b) Average Nusselt number (Nu_{avg})

	1st sphere		2nd sphere		3rd sphere	
Re	Present	ref. [7]	Present	ref. [7]	Present	ref. [7]
$\beta = 2$, $t = 2$						
5	2.65	2.67	0.74	0.75	0.21	0.217
10	3.77	3.79	1.80	1.82	0.90	0.914
20	4.79	4.82	3.03	3.05	2.11	2.13
50	6.35	6.38	4.38	4.41	3.71	3.73
100	8.03	8.08	5.41	5.44	4.76	4.78
$\beta = 6$, $t = 4$						
5	2.91	2.92	2.47	2.48	2.25	2.26
10	3.40	3.41	2.82	2.83	2.63	2.64
20	4.11	4.13	3.27	3.29	3.03	3.04
50	5.51	5.53	5.11	5.14	3.75	3.76
100	7.05	7.08	6.02	6.05	4.50	4.53

undergoing rearrangement due to the disturbance introduced by the spheres. Indeed, both these conjectures are borne out by the results seen in these figures. For $\beta = 2$, the fluid-like regions continue in the lateral direction all the way up to the wall for all values of *Re*, *Bn* and *t*. Thus, the spheres in this case should experience some wall effects, similar to that reported by Das et al.[21] and Yu and Wachs[23] for a single sphere. For $\beta = 2$, at low Bingham numbers, there are very small caps of unyielded material adhering at the stagnation points which diminish with the increasing Reynolds number, but these grow in the axial direction with the increasing Bingham number to the extent that bridges form of the unyielded material connecting spheres with each other. With the increasing Bingham number, the width of such bridges also increases. Not only this, small islands of unyielded material also appear on the sides of the bridges. Besides, the fluid yielding commences up to $z = -2$ upstream of the first sphere and it persists up to $z = 6$ downstream of the third sphere at $Re = 100$ and $Bn = 1$. The corresponding values of z (upstream and downstream) decreases with the increasing Bingham number and/or decreasing Reynolds number. In the limit of $Bn \to \infty$ ($Bn = 100$), the yield surfaces are seen to approach a limiting shape.

For $\beta = 4$ and $t = 6$ (not shown here for the sake of brevity), qualitatively similar trends are seen except for a few new features appear in the flow field. Firstly, each sphere is now surrounded by its own cavity of yielded material and these cavities are separated by unyielded material in between them. Therefore, in this case if at all there is a communication between the two spheres, it must be via the stress field. Other than this, each sphere is likely to behave as an isolated sphere. Secondly, at $Bn = 1$, the fluid yielding persists over greater axial distances than that at $\beta = 2$ and $t = 1.5$. Thirdly, depending upon the values of *Bn* and *Re*, fluid-yielding ends before reaching the confining walls and thus

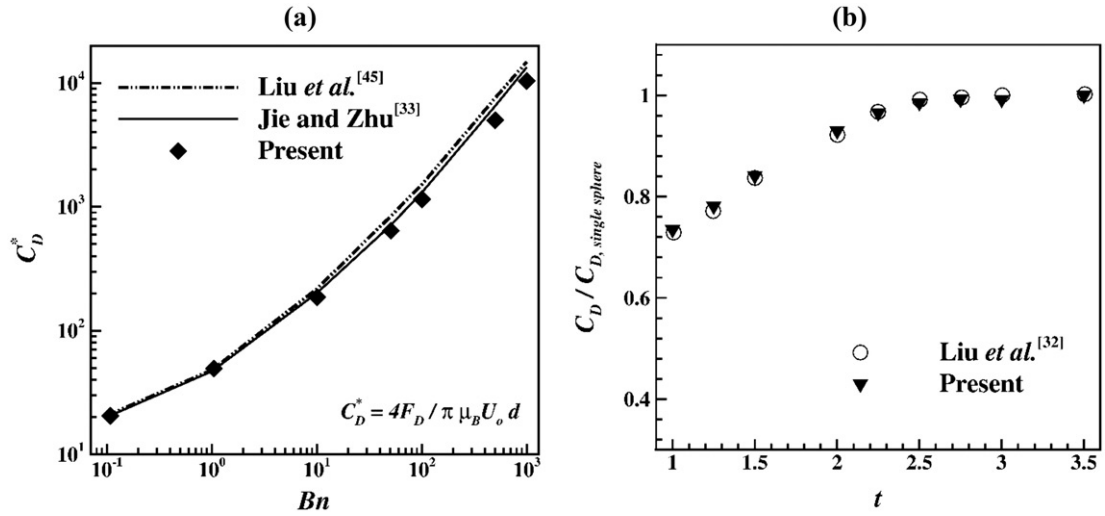


Figure 7. Comparison of the present drag results (a) single sphere at $Re = 0.1$ (b) two in-line spheres at $Bn = 340.7$.

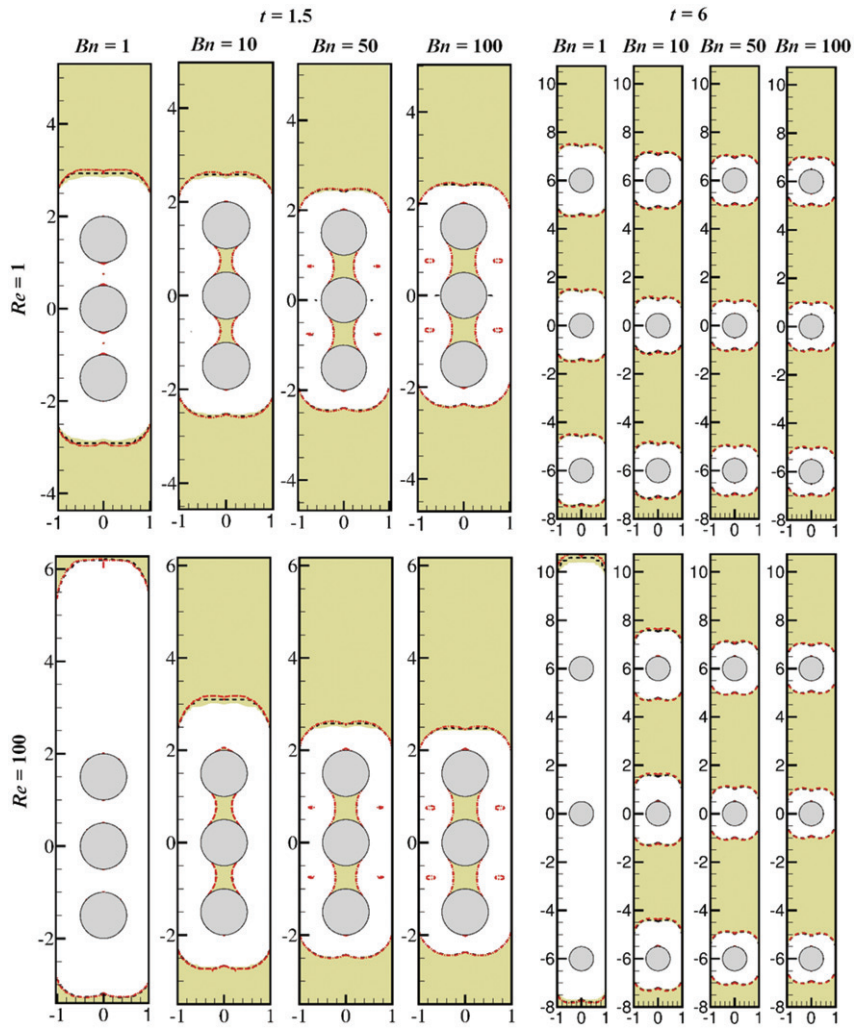


Figure 8. Yield surfaces for $\beta = 2$. Shaded: Papanastasiou model, dashed lines: bi-viscous model, red lines: Bercovier and Engelman model.

the wall effects under these conditions must be negligible. Finally, this section is concluded by noting that the predictions of the three regularization methods are seen to be in excellent agreement thereby lending credibility to the present results.

Similarly, the streamline plots were examined in detail. As expected, at low Reynolds numbers, the flow remains attached to the surface of each sphere though the velocity field impinging on the second and third sphere was modified with reference to the uniform velocity impinging on the

first sphere. No recirculating region was observed in the rear of any of the spheres at $Re = 1$ irrespective of the value of Bn, β and t. On the other hand, at $Re = 100$, there was a well-formed wake at all values of β and t, but only at small Bingham numbers, i.e., when the fluid behaviour is close to the Newtonian behaviour limit. This tendency was, however, suppressed completely at high Bingham numbers. Naturally, the severity of interactions between spheres is dependent on the fact that whether the wake region extends up to the trailing sphere or not. Hence, such interactions weaken with the increasing value of t. Also, suffice it to add here that the confinement also imparts stability to the flow by deferring flow separation to higher Reynolds number with the decreasing value of β. However, no attempt was made in this work to establish the critical values of the Reynolds number for the onset of wake formation or of the Bingham number beyond which no wake would be formed for given values of Re, β, and t.

4.3. Isotherm contours

Figure 9 shows representative isotherm contours for varying blockage ratios and sphere-to-sphere distances for different values of Re, Pr and Bn. Isotherm contours near the surface of the spheres are shown for $Re = 1$ on the left-hand side and $Re = 100$ on the right-hand side. In the yielded regions, heat transfer occurs primarily by convection whereas in the solid-like unyielded regions, conduction dominates. At low Peclet numbers ($Re.Pr$), due to weak advection effects, the isotherms tend to spread far and wide (low temperature gradient) and form nearly concentric circles at the front of the spheres. With increasing Bn for a fixed blockage ratio (β) and sphere-to-sphere distance (t), the density of isotherms increases near the surface of each sphere. This is due to the shrinking yielded regions, which sharpens the velocity and temperature gradients in the close proximity of each sphere. Furthermore, as the gap between the two spheres increases, similar trends are seen in isotherms. In summary, due to the increasing blockage effect, the thermal boundary layer thickness is restricted. This can influence the rate of heat transfer in a positive or negative sense depending upon whether the fluid yielding extends all the way up to the tube wall or not.

4.4. Total drag coefficient

Due to the prevailing tangential and normal stresses acting on the surface of each sphere, the fluid exerts a net hydrodynamic drag force on each sphere, which is made up of two components, namely, form drag (C_{DP}) and friction drag (C_{DF}). Evidently, both the individual and total drag coefficients are expected to be functions of Re, Bn, t and β. Representative results elucidating this functional relationship are shown in Figure 10. Based on a detailed examination of the entire data set, the key trends can be summarized as follows: the drag coefficient of each sphere shows a positive dependence on Bn and t while a classic inverse dependence on Re. Thus, the drag coefficient for each sphere is higher in Bingham plastic fluids than that observed in the Newtonian fluids under otherwise identical conditions which is evident from the value of $C_D/C_{D(Bn=0)} > 1$ as seen in Figure 10(a) irrespective of the value of Bn. Naturally, the smaller is the value of t, the stronger is the interaction between two neighbouring spheres. As the gap between the two spheres increases, each sphere behaves increasingly as an independent sphere. The 1st sphere always experiences a higher drag than the 2nd and 3rd spheres since the 2nd and 3rd spheres are exposed to a velocity profile disturbed by the first sphere which results in a reduced drag due to the reduced velocity gradients. This effect is almost negligible at lower values of Re due to weak inertial effects and because the flow remains attached. The ratio $C_D/C_{D(Bn=0)}$ is equal for the 1st and 3rd sphere at $Re = 1$ due to the identical existence of unyielded material around the 1st and 3rd sphere; while this ratio is lowest for the 1st sphere at high Re (i.e., $Re = 100$), Figure 10(a). Negligible effects of the gap between two consecutive spheres (value of t) on drag were observed for the 1st sphere whereas lower values of t resulted in higher drag on the other two spheres as can be seen in Figure 10(b). In Newtonian fluids, this is due to the wake formation which may extend all the way up to the next sphere. In Bingham plastic fluids, such interference is further accentuated by the presence of unyielded material at the stagnation points. For example, at $Re = 100$ and $Bn = 100$, the increase in drag ratio is ~15-20% for 2nd and 3rd sphere when t is reduced from 6 to 1.5. This is simply due to the increased hydrodynamic interactions between the flow fields in the proximity of the spheres at $t = 1.5$ that that seen at $t = 6$. Also, the wall effects play a significant role in modulating the total drag coefficient. As the confinement is increased (value of β is decreased), there is an extra retardation effect caused by the tube walls thereby increasing the drag on each sphere. In Bingham plastic fluids, the ratio $C_D/C_{D(Bn=0)}$ was always found to be smaller for low values of β than that observed for large β, see Figure 10(c). For instance, at $Re = 100$ and $Bn = 100$, the increase in the value of $C_D/C_{D(Bn=0)}$ is ~100%, ~120% and ~150% for the 1st, 2nd and 3rd sphere, respectively, when confinement is reduced by 50%, i.e., β is increased from 2 to 4, as seen in Figure 10(c). Qualitatively similar trends are also obtained for the variation of pressure drag coefficient (C_{DP}) with Re, Bn, t and β (not shown here for the sake of brevity). This suggests the wall effects to be less severe in viscoplastic fluids than that in Newtonian fluids which is in line the scant experimental results available for a single sphere.[22,46] Next, the drag ratio (pressure to friction drag) was examined for a range of values of Bn and Re and for the two extremes of t and β. It is worthwhile to recall here that for an unconfined single sphere in a Newtonian fluid (creeping flow), this ratio is 0.5. For the present case and irrespective of the values of Re, Bn, t and β, the ratio C_{DP}/C_{DF} is found to be always >0.5 and as expected it further increases with Re thereby indicating a faster increase in the form drag due to the diminishing role of viscous effects. In addition, this ratio bears a positive dependence on Bn; however, at high values of Bn, the effect of Re is rather small. Finally, in the limit

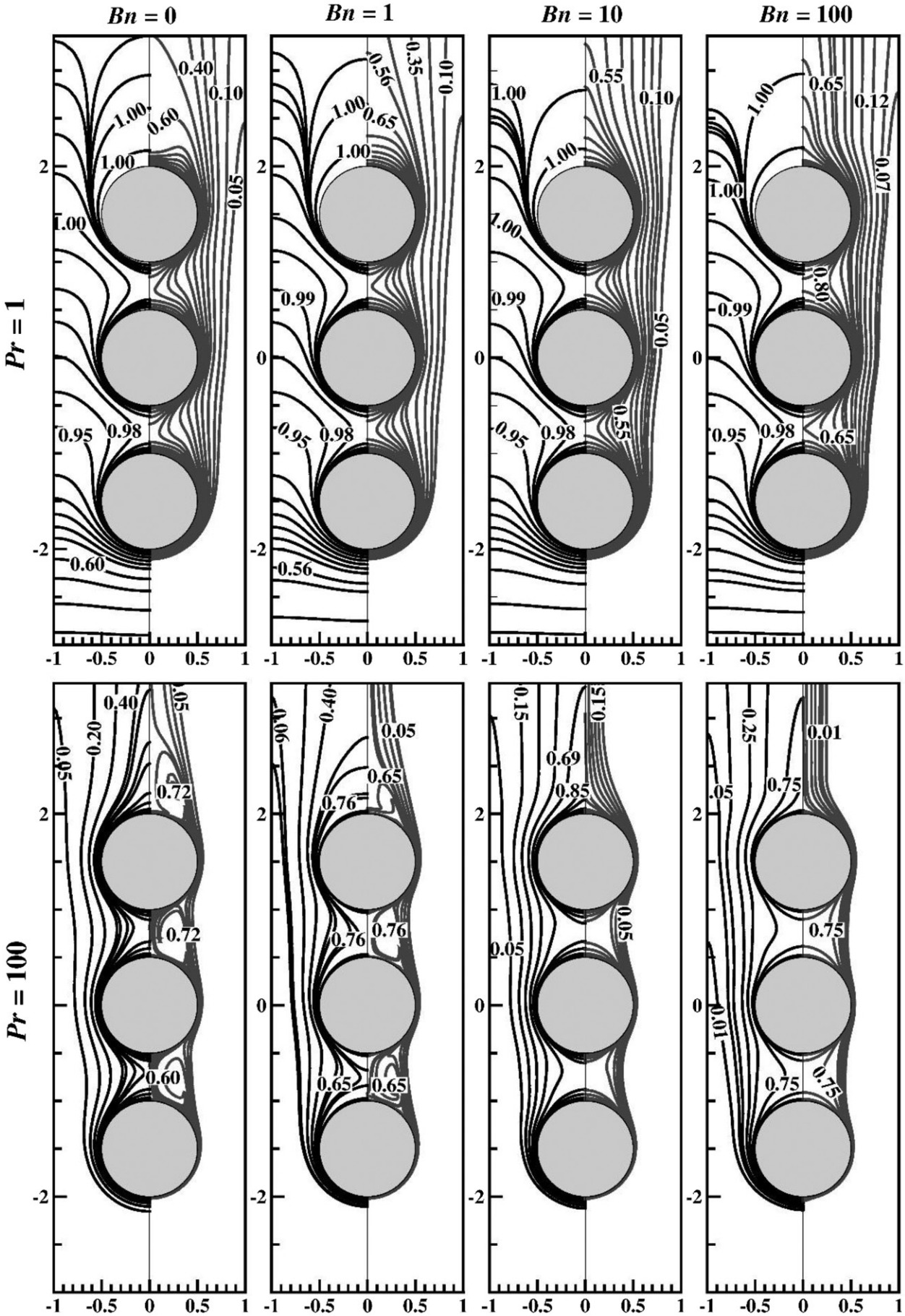


Figure 9. Representative isotherm contours for $t = 1.5$, $\beta = 2$. Left half of each sphere: $Re = 1$ and right half of each sphere: $Re = 100$.

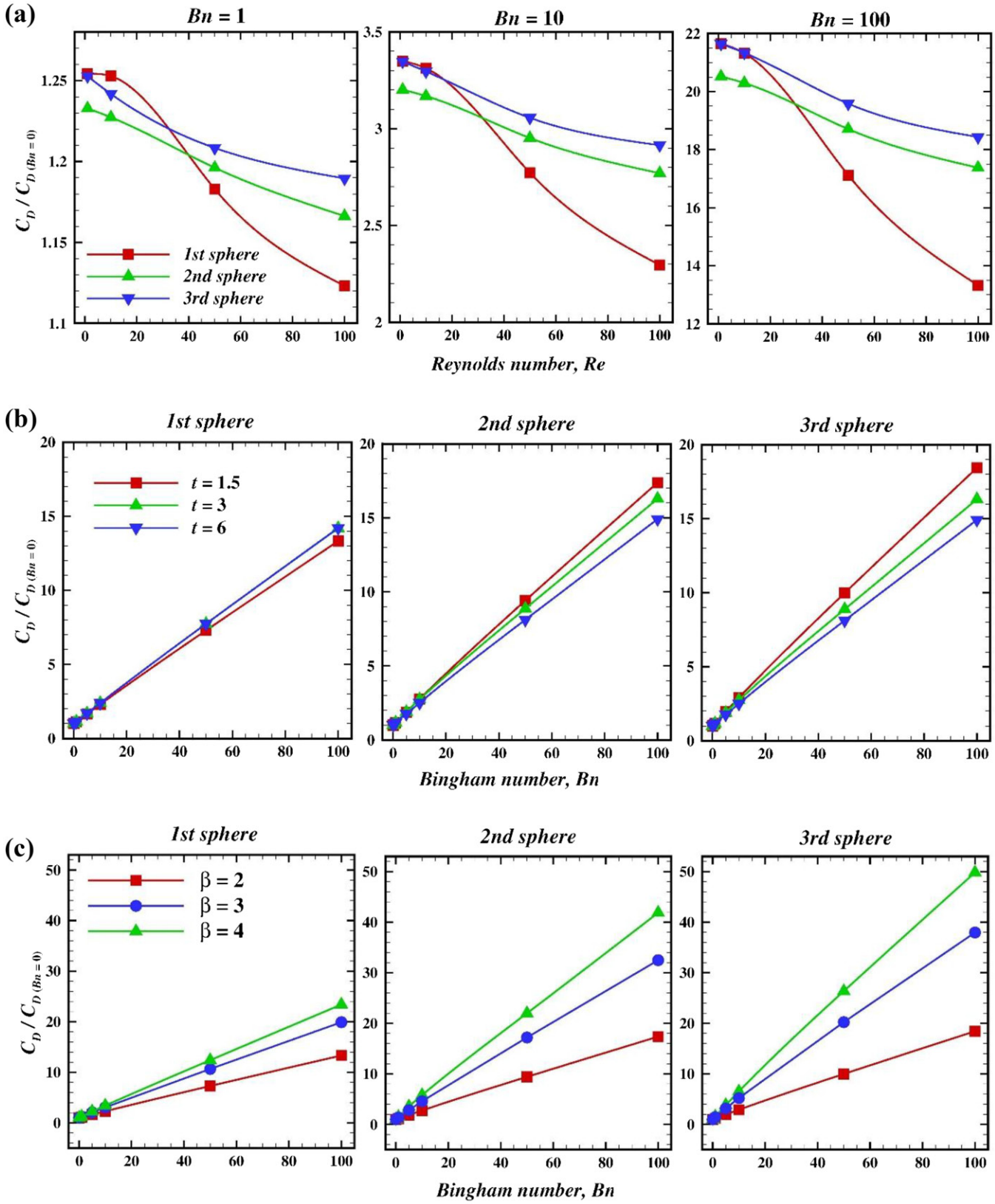


Figure 10. (a) Effect of *Re* and *Bn* on $C_D/C_{D(Bn=0)}$ for each sphere at $\beta = 2$, $t = 1.5$; (b) Effect of t on $C_D/C_{D(Bn=0)}$ at $\beta = 2$ and $Re = 100$; (c) Effect of β on $C_D/C_{D(Bn=0)}$ at $t = 1.5$ and $Re = 100$.

$Bn \rightarrow \infty$ (i.e., $Bn = 100$ here), $C_{DP}/C_{DF} \sim 2.25$ which is consistent with the fully plastic flow conditions, in line with similar trends reported in the literature for a single unconfined sphere.[19,26]

4.5. Local Nusselt number

Figure 11 shows the combined effects of *Re*, *Pr* and *Bn* on the variation of the local Nusselt number along the surface of each sphere for $t = 1.5$ and $\beta = 2$. At low *Re* and *Pr*, the

value of Nu_θ along the surface of each sphere varies very little. Also, the positive effect of Bingham number is seen to manifest itself with the increasing Peclet number (i.e., see the results for $Re = 1$ and $Pr = 100$). Also, the results for the 1st sphere are similar to that for an isolated sphere whereas the distribution of Nu_θ on the surface of each successive sphere is influenced by the modified velocity and temperature fields. Due to the preheating of the fluid, the driving force for heat transfer progressively decreases from the first sphere onwards. The lowering of the Nusselt number can be safely, at least in part, ascribed to this effect. Other than this, both Reynolds number and Prandtl number are seen to exert a positive influence on the value of the Nusselt number. With the increasing advection (with the increasing Re or Pr or both), the positive effect of Bn can be attributed to the sharpening of the velocity and temperature gradients, especially in the narrow gap between the sphere and the tube at $\theta = 90^\circ$. On both sides of this point, due to the diverging flow areas, the velocity gradient decreases thereby lowering the prevailing stress levels which promotes the formation of unyielded regions. This lowers the local Nusselt number both in the front half and rear half of the sphere. Both these effects weaken with the increasing values of t and β. The nature of the local Nusselt number distribution for each sphere is closer to that for a single sphere than that seen in Figure 11. Unlike in the case of a Newtonian fluid where the maximum value of the Nusselt number occurs at $\theta = 0^\circ$ (front stagnation point), in the present case for $Bn > 0$, the local Nusselt number no longer is maximum at the front stagnation point rather it shifts a bit away from it. This is due to the unyielded polar caps present at the front stagnation point. All in all, the local Nusselt number shows a positive dependence on Re, Pr and Bn. Also, the rate of heat transfer is further augmented with the increasing values of t due to the increased level of shearing near the surface of the sphere leading to the sharpening of velocity gradients. Furthermore, with the decreasing value of β (increasing confinement), the value of theNu_θ is increased due to the sharpening of the temperature gradients near the surface of the spheres and/or due to the fluid acceleration thereby leading to higher values of the local Reynolds number than those based on the global characteristics.

4.6. Average Nusselt number

While the local Nusselt number plots yield detailed information about the relative contribution to the overall rate of heat transfer from different parts of the heated sphere, the surface averaged Nusselt number values are frequently required in process engineering and design calculations. In order to elucidate the effect of fluid yield stress in an unambiguous manner, the variation of the ratio $Nu_{avg}/Nu_{avg(Bn=0)}$ with Re, Pr, Bn, β and t is shown in Figure 12 for a range of conditions. In most cases, this ratio is greater than unity (except for $Bn = 1$ and for 2nd and 3rd sphere) thereby showing an augmentation in heat transfer over and above that in Newtonian fluids. Furthermore, this ratio shows a positive dependence on Re, Bn and Pr due to the thinning of the momentum and thermal boundary layers. At low values of Bn and at high Re, Figure 12(a), the ratio $Nu_{avg}/Nu_{avg(Bn=0)}$ is seen to fall below unity for 2nd and 3rd sphere. This possibly may be due to weak yield stress effects and at $Re = 100$ there will be a well-formed wake in the rear of the 1st sphere which extends up to the 2nd sphere thereby lowering the heat transfer from the 2nd and 3rd sphere at such small values of $t = 1.5$. Overall, at low values of Re, the dependence of Nu_{avg} on Pr and Bn is weak due to feeble convection effects (conduction dominant). On the contrary, at high Re, the Prandtl number exerts a significant impact on the rate of heat transfer. Irrespective of the values of Re, Nu_{avg} shows a positive dependence on Bn at higher Pr, but remains constant at low Pr due to the dominant role of conduction with decreasing Pr. The dependence of Nu_{avg} on Bn can be explained in the following way: with the increasing Bingham numbers, the velocity gradients become large in the thin yielded regions attached to the surface of the sphere, thereby increasing the overall rate of heat transfer. Figure 12(b) shows that the effect of the gap between the spheres diminishes with the increasing value of t. For instance, the smaller values of t yield better heat transfer for the 3rd sphere while reverse effects were seen to exist for the 1st sphere. Moreover, irrespective of the values of all other parameters, the larger values of β (low confinement) increase the rate of heat transfer over and above that seen in their Newtonian counter parts for each sphere, Figure 12(c). This is due to the fact that the walls are adiabatic and therefore, farther are the walls from the sphere, higher is the heat transfer. Altogether, at fixed values of Re, Bn and Pr, the Nusselt number is seen to be maximum for the 1st sphere and minimum for the 3rd sphere.

From dimensional analysis, the average Nusselt number is expected to be a function of Re, Pr, Bn, t and β, i.e.,

$$Nu_{avg} = f(Re, Pr, Bn, t, \beta) \quad [19]$$

By employing the modified definitions of Re^* and Pr^* given in Equations (12) and (13), the above functional form can be reduced as:

$$Nu_{avg} = f(Re^*, Pr^*, t, \beta) \quad [20]$$

Further consolidation of the results for different values of Pr^* can be achieved by introducing the familiar Colburn factor, j_H, as:

$$j_H = \frac{Nu_{avg}}{Re^* Pr^{*1/3}} = f(Re^*, t, \beta) \quad [21]$$

For the range of conditions considered herein, the final form of Equation (21) was obtained as follows:

$$j_H = a' \cdot \frac{(t+1)^{b'}}{\left(1+\frac{1}{\beta}\right)^{c'}} \cdot Re^{*-2/3} \quad [22]$$

The values of the fitting constants in Equation (22) are summarized in Table 3 individually for the three spheres. Equation (22) reproduces 720 numerical data points with the mean and maximum deviations listed in Table 3. While the relatively large deviations between the actual data and the predictions of Equation (22) are discomforting, the fact that no such expression exists even for Newtonian fluids

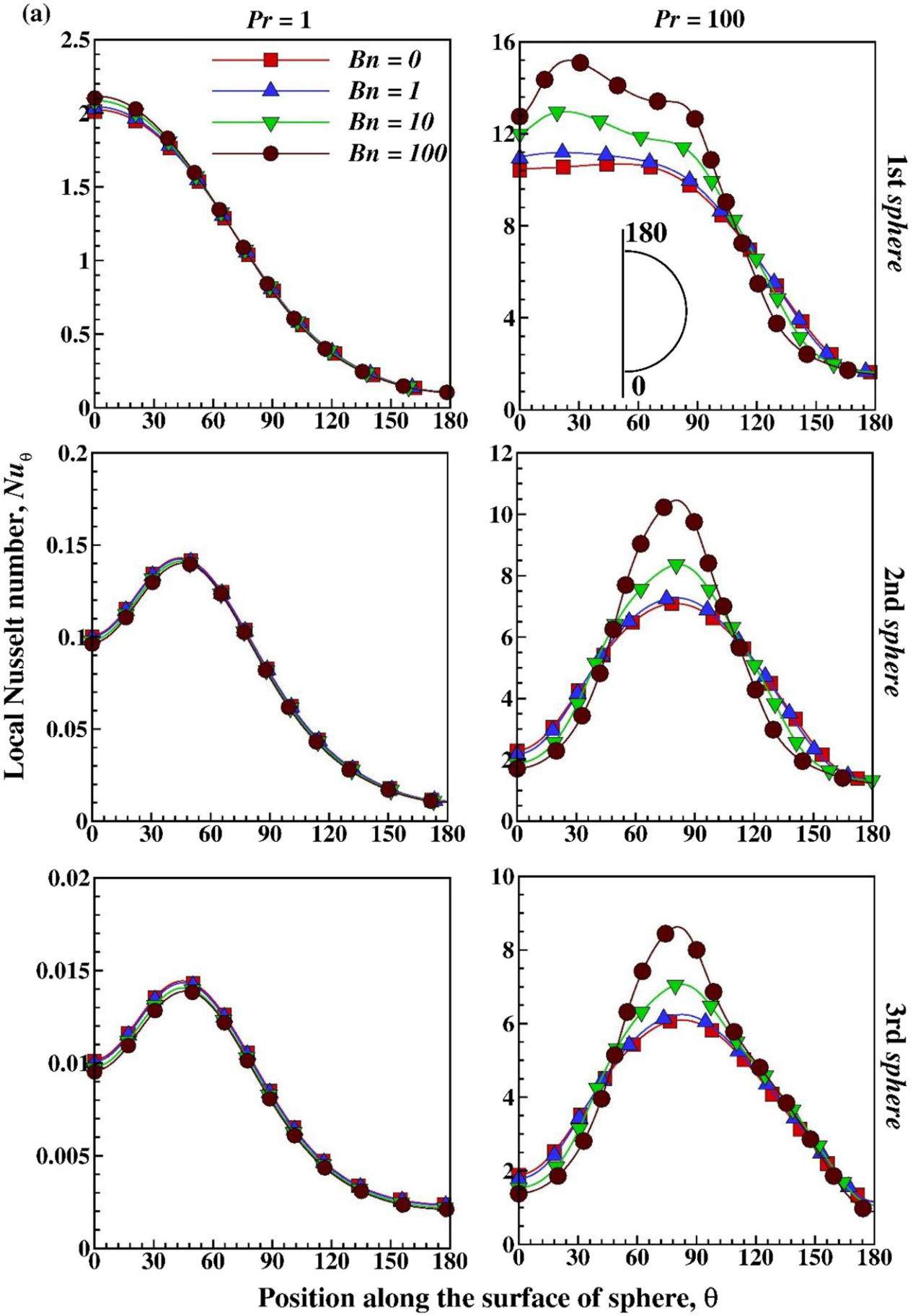


Figure 11. Variation of the local Nusselt number along the surface of each sphere for $t = 1.5$, $\beta = 2$ (a) $Re = 1$ (b) $Re = 100$.

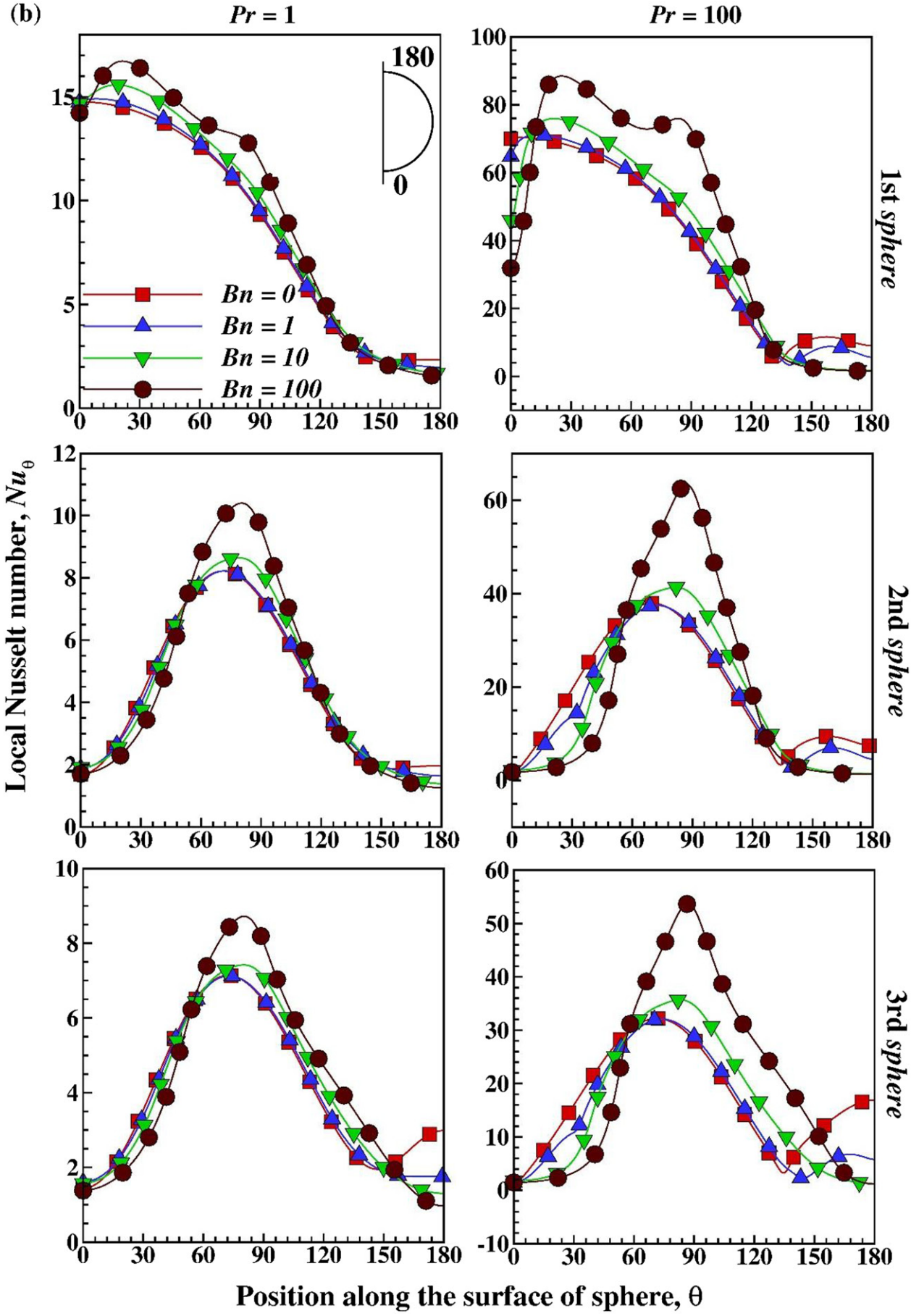


Figure 11. (Continued)

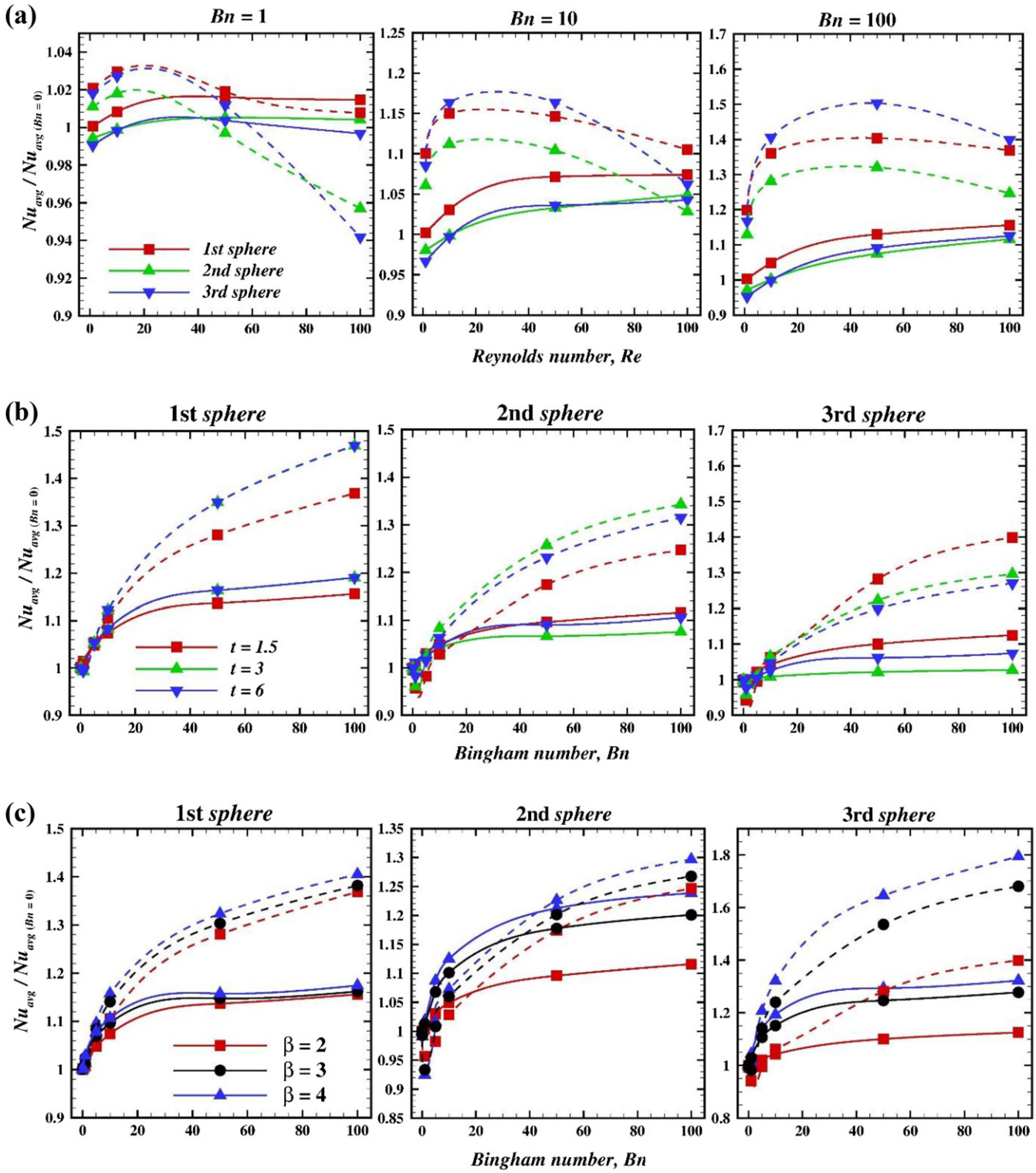


Figure 12. (a) Effect of *Re* and *Bn* on $Nu_{avg}/Nu_{avg(Bn=0)}$ for each sphere at $\beta = 2$, $t = 1.5$; (b) Effect of t on $Nu_{avg}/Nu_{avg(Bn=0)}$ at $\beta = 2$ and $Re = 100$; (c) Effect of β on $Nu_{avg}/Nu_{avg(Bn=0)}$ at $t = 1.5$ and $Re = 100$. Solid lines: $Pr = 1$, dashed lines: $Pr = 100$.

Table 3. Values of fitting constants used in Equation (22).

				% Error	
	a'	b'	c'	Mean	Max
1st sphere	1.674	0.069	−0.364	12.4	34.9
2nd sphere	0.895	0.247	−0.051	13.6	39.7
3rd sphere	0.786	0.364	0.369	15.2	46.9

testifies to the intrinsic difficulties in devising such expressions. Further analysis of the results showed that only ~8% of data points showed deviations greater than 25% without any discernible trends. Also, other forms of functions were attempted but without any significant improvement in the degree of fit.

5. Conclusions

This work deals with the flow past and heat transfer from an in-line array of three spheres fixed on the axis of a cylinder with the uniform flow of a Bingham plastic fluid. Extensive results on streamline and isotherm contours, yield surfaces, drag coefficients, and local and average Nusselt number are discussed over wide ranges of conditions as: $1 \leq Re \leq 100$, $1 \leq Pr \leq 100$, $0 \leq Bn \leq 100$, $1.5 \leq t \leq 6$ and $2 \leq \beta \leq 4$. Due to the fluid yield stress, the flow domain is spanned by fluid-like (yielded) and solid-like (unyielded) sub-regions. The yielded regions are suppressed with the increasing Bn values together with the reduced tendency of flow separation. This influences both the drag and heat transfer. Overall, the drag coefficient shows a positive dependence on Bn and t while an inverse dependence on Re and β. All else being equal, the values of the Nusselt number in Bingham plastic fluids are larger than that in Newtonian fluids and this gets accentuated with the increasing values of Re, Pr, Bn and β. Finally, a compact functional relationship between the average Nusselt number is presented in terms of the modified Reynolds (Re^*) and Prandtl (Pr^*) numbers, t and β. This can be used to estimate the value of the average Nusselt number in a new application.

Nomenclature

Bn	Bingham number, $(Bn = \tau_0 d/\mu_B U_0)$, dimensionless;
Bn^*	modified Bingham number, $Bn^* = (Bn/(1+Bn))$, dimensionless;
C	thermal heat capacity of fluid, J.kg^{-1}.K^{-1};
CD	drag coefficient, dimensionless;
C_{DF}	friction drag coefficient, dimensionless;
C_{DP}	pressure drag coefficient, dimensionless;
C_p	pressure coefficient, $C_p = 2(p_s - p_\infty)/\rho U_0^2$, dimensionless;
d	diameter of sphere, m;
D	diameter of the tube, m;
h	heat transfer coefficient, W.m^{-2}.K^{-1};
j_H	Colburn j-factor, dimensionless;
L_d	downstream length, dimensionless;
L_u	upstream length, dimensionless;
m	growth rate parameter used in Papanastasiou model, dimensionless;
Nu_{avg}	average Nusselt number, dimensionless;
Nu_θ	local Nusselt number, dimensionless;
P	pressure, dimensionless;
p_s	local pressure on the surface of the sphere, Pa;
p_∞	reference pressure far away from the sphere, Pa;
Pr	Prandtl number, dimensionless;
Pr^*	modified Prandtl number, $Pr^* = (Pr(1+Bn))$, dimensionless;
R	radius of sphere, m;
Re	Reynolds number, dimensionless;
Re^*	modified Reynolds number, $Re^* = (Re/(1+Bn))$, dimensionless;
s	sphere-to-sphere distance, m;
t	sphere-to-sphere distance, $t = s/d$, dimensionless;
T	temperature of fluid, dimensionless;
T'	temperature of fluid, K;
T_w	sphere surface temperature, K;
T_0	free stream temperature, K;
$\mathbf{V}$	velocity vector, dimensionless;
U_0	uniform inlet velocity, m.s^{-1};
V_r	velocity component in r-direction, dimensionless;
V_z	velocity component in z-direction, dimensionless;
r, z	cylindrical coordinates, dimensionless; Greek symbols;
β	blockage ratio, $\beta = D/d$, dimensionless;
$\dot{\gamma}$	rate of strain tensor, dimensionless;
η	scalar viscosity function, dimensionless;
θ	position on the surface of the sphere, deg;
λ	regularization parameter in Bercovier and Engelman model, dimensionless;
μ_B	Bingham plastic viscosity, Pa.s;
μ_y	yielding viscosity, Pa.s;
ρ	density of the fluid, kg.m^{-3};
τ	extra stress tensor, dimensionless;
τ_0	yield stress, Pa;
∇	del operator, dimensionless;

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